Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Enterprise solutions engineer with 2+ years supporting **education, healthcare, and communication systems**; experienced translating customer needs into technical designs, validating integrations, coordinating deployments, troubleshooting production issues, documenting requirements, and delivering reliable service.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Worked with stakeholders to **assess needs and define solution requirements** for an education platform serving 500+ users, designing data structures and integrations while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Validated application and integration behavior across 5,000+ monthly workflows, documenting requirements, errors, reconciliation results, and corrective actions while coordinating tested CI/CD releases that maintained 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Translated instructor requirements into **education application designs** for uploads, processing, history, and reporting, delivering React interfaces and Flask services that supported 150+ users and 10,000+ daily requests.
- Provided technical support by reproducing file and integration issues, validating solutions in development environments, and implementing fallback handling, automated tests, and documented fixes that reduced user turnaround time 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Supported **system design, integration, and deployment** across Windows, Linux, and macOS, coordinating requirements with Microsoft and GitHub stakeholders while improving API response times 30% and pipeline efficiency 68%.
- Built laboratory-style validation workflows for installation and service changes, automating executable builds, configuration checks, failure reports, and 20+ unit and integration tests to produce controlled, repeatable releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Designed and integrated **real-time communication and information services** across eight components and 2M+ records, supporting authentication, resource assignment, permissions, notifications, reporting, and dashboards through WebSockets and Azure SQL.
- Collaborated with project and deployment teams to validate integration behavior, monitor system health, troubleshoot production issues, and release tested Docker configurations through Jenkins while reducing deployment time 50%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Partnered directly with clients to **analyze requirements and present system designs**, mapping 12+ integration points and documenting end-to-end workflows before implementation, reducing design errors 25% and accelerating delivery 22%.
- Supported deployment and ongoing service for a cross-platform application covering bookings, pricing, payments, and status updates across 50,000+ records, troubleshooting integrations while maintaining sub-200-millisecond responses and improving compatibility 15%.

Skills

Solutions: Needs assessment, requirements validation, system integration, configuration, deployment, service support
Design & Documentation: Figma, system architecture, workflow diagrams, technical specifications, validation records
Communication Systems: REST APIs, WebSockets, real-time notifications, monitoring, access control, reporting
Technology: Python, Java, JavaScript, TypeScript, SQL Server, Azure SQL, PostgreSQL, Windows, Linux
Quality & Delivery: Integration testing, development environments, troubleshooting, Git, Jenkins, CI/CD, Docker, Agile

Projects

HMIS Live | *Healthcare Systems, REST APIs, Sentry*

- Designed a **healthcare information and workflow system** with consent-based patient access, appointment coordination, dashboards, notifications, REST integrations, and Sentry monitoring, halving record retrieval time while reducing administrative workload 30%.

IoT Authentication Analysis | *IoT, Device Security, Access Control*

- Designed and validated **secure IoT device authentication** across three user scenarios and four permission tiers, documenting integration assumptions and replacing centralized verification with tamper-resistant identity workflows while reducing repeated key checks 50%.

Chat2SQL Code | *System Integration, SQL Server, Docker*

- Enhanced a **cross-platform integration and database application** supporting SQL Server and 10+ data systems with schema exploration, compatibility checks, query generation, Docker configurations, and reusable development and deployment workflows.